SDLC Model

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For the Micro-Service Hub Platform, a comparative analysis of different SDLC models was conducted based on selected project criteria. Each criterion was assigned a priority according to its importance to the system. The models were then evaluated against these criteria, and scores were calculated accordingly. It is given in Table 1.

Table 1: Comparison matrix with different models

Priority	Criteria	Waterfall	V-shape	Iterative	Spiral	Agile	Prototype
5	Dynamic/Evolving requirements	No	No	Yes	Yes	Yes	Yes
3	Technological knowledge	Yes	Yes	No	No	No	No
6	Efficiency & Rapid Delivery	No	No	Yes	No	Yes	Yes
3	Risk analysis	No	No	No	Yes	Yes	No
5	User testing ability	No	No	Yes	Yes	Yes	Yes
5	Dependability and Security	Yes	Yes	No	Yes	Yes	No
3	Time and Cost Effective	No	No	Yes	No	Yes	Yes
Total- 30	Over all	8	8	19	18	27	19

Justification: The results are as follows: Waterfall (8), V-Shaped (8), Iterative (19), Spiral (18), Agile (27), and Prototype (19). The Agile model achieved the highest score, indicating strong support for critical project requirements such as handling dynamic requirements, ensuring rapid delivery, and facilitating continuous user testing. Given that a service hub platform requires high adaptability for two distinct user roles (Seekers and Providers) and continuous validation of features like payment gateways and messaging, the Agile model is the most suitable choice. Its iterative sprint approach ensures early detection of integration errors and enhances overall user satisfaction through constant feedback.